A Chebyshev-Leaver Spectral Residual Workflow for Kazakov-Solodukhin Quasinormal Modes

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Abstract. We compute low-lying quasinormal modes (QNMs) of Kazakov–Solodukhin (KS) black holes with an audited Chebyshev–Leaver spectral residual workflow. Raw generalized eigenvalues are accepted only after smallest-singular-value, spectral-size, continuation, backward-error, and collocation-independent continued-fraction checks. For the Schwarzschild scalar $\ell = 2, n = 0$ mode, the $N = 96$ Chebyshev result and the continued-fraction result agree with an independently published frequency to relative differences 5.60×10^{-11} and 8.24×10^{-13}, respectively. At fixed mass, the real frequency and damping magnitude of the robust scalar fundamentals decrease successively across the four sampled values $a/M = 0, 0.2, 0.5, 1$. For $\ell = 2, n = 0$ at $a/M = 1$, the oscillation frequency, damping magnitude, and quality factor $Q = \mathrm{Re}\,\omega/[2(-\mathrm{Im}\,\omega)]$ shift by -7.29%, -2.59%, and -4.83%. The low-lying catalogue contains scalar $\ell = 2, 3, 4$ fundamentals, cross-validated first overtones, and exploratory second-overtone diagnostics; higher overtones are outside the scope. A finite-N singular-value pseudospectral diagnostic of the scalar $\ell = 2, n = 0$ branch broadens with deformation: at $N = 64$, its 10% quantile diagnostic increases, with a positive endpoint change under the reported grid, window, and spectral-size robustness checks. The odd-parity gravitational sector is gauge invariant within an explicitly added inverse-Cowling closure for the effective KS stress tensor. Its QNMs are conditional on that closure, because the spherical KS effective action does not determine independent nonspherical quantum-source perturbations.

Keywords. Kazakov-Solodukhin black holes; quasinormal modes; Chebyshev spectral methods; Leaver continued fractions; spectral residual minimization; black-hole spectroscopy.

1 Introduction

The ringdown phase of a perturbed black hole is governed by a discrete set of complex resonant frequencies known as quasinormal modes (QNMs). In the classical picture, these frequencies depend only on the parameters of the final black hole and on the field or metric perturbation being considered. Their real parts set the oscillation frequencies, while their imaginary parts determine damping or instability. QNMs have therefore become a privileged language for connecting mathematical relativity, black-hole stability, and gravitational-wave astronomy [8, 9, 10].

The first direct detection of gravitational waves from a binary black-hole merger established ringdown as an observationally relevant strong-field regime [11]. Subsequent black-hole spectroscopy studies have used ringdown harmonics and overtones to test the Kerr hypothesis and the no-hair theorem [12, 13]. These developments motivate increasingly precise calculations of QNM spectra not only in general relativity, but also in quantum-corrected, modified-gravity, and exotic compact-object backgrounds.

QNM frequencies are obtained with several complementary tools. WKB approximations connect the spectrum to the height and curvature of the effective-potential barrier and are

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efficient when the multipole number exceeds the overtone number [19, 20, 10]. Direct time-domain evolution extracts the least-damped modes from waveforms, while Frobenius/Leaver continued fractions impose the frequency-domain boundary conditions through a minimal-solution recurrence [18]. The asymptotic iteration method provides another recursive treatment of the radial ordinary differential equation [21]. Spectral and pseudospectral methods instead discretize the boundary-value problem globally and can expose many candidate eigenvalues at once [22]. The present workflow combines the last approach with time-domain orientation, continued-fraction cross-validation, and singular-value diagnostics; it does not replace these established methods.

The Kazakov–Solodukhin (KS) metric is a quantum-deformed Schwarzschild geometry originally derived from an effective two-dimensional dilaton-gravity reduction of spherically symmetric quantum fluctuations [14]. It modifies the Schwarzschild interior by replacing the classical center with a finite-radius structure and has since been used as a phenomenological model for quantum-corrected black-hole physics [15, 16, 17]. Recent QNM studies of the KS spacetime suggest that fundamental modes may remain close to Schwarzschild values while overtones and evaporation-related quantities can be more sensitive to quantum deformation [16, 17]. This makes the model a useful target for computational experiments: it is close enough to Schwarzschild to permit strong benchmarking, yet different enough to probe quantum-gravity phenomenology.

At the same time, QNM computation is intrinsically a non-Hermitian resonance problem. The boundary conditions are outgoing or ingoing rather than self-adjoint, the frequencies are complex, and the numerical spectrum can be sensitive to discretization and operator perturbations. The methodological question asked here is therefore not only "what are the KS QNM frequencies?" but also "can non-Hermitian QNM resonance conditions be audited through Hermitian residual minimization?" The paper answers this question operationally: construct the spectral QNM operator $P_N(\omega)$, form $R_N(\omega) = P_N(\omega)^\dagger P_N(\omega)$, search for minima of $\sigma_{\min}(P_N(\omega))$, and validate the resulting branches against convergence and a Leaver-style discretization that is independent of Chebyshev collocation, matrix-pencil data, and residual minimization while sharing the same perturbation equation and endpoint factorization.

This framing should be read carefully. Spectral, pseudospectral, and pseudospectrum methods already provide powerful QNM tools, and black-hole pseudospectrum studies explicitly connect non-self-adjoint QNM operators, Chebyshev discretizations, and singular-value diagnostics [22, 26, 27]. Smallest-singular-value methods are also established in the broader nonlinear eigenvalue literature [28]. This manuscript does not claim priority for singular-value, pseudospectrum, or residual formulations. Its goal is to implement, audit, and apply a concrete Chebyshev–Leaver spectral residual workflow for KS QNMs, then separate the resulting workflow, numerical, and physics claims.

The contribution is not a new high-precision eigensolver or a general solution to the high-overtone problem. It is an audited workflow in which raw generalized-eigenvalue roots are treated as candidates and accepted only after residual, convergence, continuation, backward-error, and collocation-independent continued-fraction checks. The resulting advantage is confidence and traceability: each reported branch carries evidence explaining why it was retained and what level of interpretation it supports.

The scope is therefore deliberately limited to a low-lying KS catalogue. The scalar $n = 0$ fundamentals are robust fundamental modes; scalar $n = 1$ branches are cross-validated low-lying first overtones at $N = 32$; $n = 2$ entries are exploratory branch diagnostics; and higher overtones are outside the scope. Recent analytic-continuation methods based on confluent-Heun solutions explicitly target complete and highly damped spectra of type-D black holes [30]. That is a different objective from the audited low-lying KS branches studied here. The scalar $\ell = 2$ fundamental supplies the physical headline. We also derive a gauge-invariant odd-parity master equation under a frozen-source closure. This is stronger than a lapse-substitution ansatz, while remaining conditional on excluding independent axial excitations of the effective quantum source.

1.1 Literature context and related work

1.1.1 Quasinormal modes and black-hole spectroscopy

The theory of black-hole perturbations begins with the Regge–Wheeler analysis of Schwarzschild perturbations [1] and early numerical recognition of damped black-hole ringing [7]. Modern reviews emphasize that QNM spectra provide information about stability, boundary conditions, late-time behavior, and possible observational tests of general relativity [8, 9, 10]. In gravitational-wave astronomy, QNM spectroscopy is especially important because detecting multiple ringdown tones can test whether the remnant is described by the Kerr family [12, 13].

1.1.2 Quantum-corrected Schwarzschild geometries

Kazakov and Solodukhin proposed a quantum deformation of the Schwarzschild solution in which the radial center is displaced to a Planck-scale surface through a spherically symmetric quantum correction [14]. Later geometric analyses have clarified that the original construction improves some regularity properties but does not automatically remove all curvature singular behavior in the sense usually meant in general relativity [15]. Phenomenological work on KS black holes has studied QNMs, scattering, shadows, Hawking radiation, and overtone sensitivity [16, 17]. These studies motivate higher-accuracy spectral calculations and systematic comparisons across deformation parameters.

1.1.3 Spectral approaches to QNM computation

Spectral methods are attractive for QNM problems because they approximate smooth functions globally and can converge rapidly when the boundary conditions and coordinate compactifications are chosen well. Leaver's Frobenius continued-fraction method remains a standard benchmark for black-hole QNM calculations [18]. Jansen's QNM spectral package demonstrates how differential QNM equations can be discretized and solved as generalized eigenvalue problems over a wide class of asymptotics [22]. More recent studies by Batic, Dutykh, and collaborators use spectral formulations to revisit noncommutative Schwarzschild black holes, Lee–Wick black holes, and Morris–Thorne wormholes, correcting or sharpening conclusions obtained from lower-order approximation methods [23, 24, 25]. Pseudospectral and pseudospectrum analyses also highlight that QNM spectra can be sensitive to operator perturbations, an important caution for finite-resolution spectral discretizations [26]. Cao et al. recently applied this pseudospectrum perspective to a quantum-corrected Schwarzschild black hole, which is especially relevant background for the present KS calculation [27].

1.1.4 Residual and singular-value viewpoints

The residual formulation used here belongs to a broader numerical landscape rather than standing outside it. For a nonlinear spectral operator $P(\omega)$, exact QNMs satisfy $P(\omega)u = 0$ for a nonzero state u. At finite resolution this condition can be tested by the smallest singular value of the discretized operator, and the associated Hermitian residual $P^\dagger P$ converts the defect into a positive-semidefinite object. Smallest-singular-value methods for nonlinear matrix eigenvalue problems are established independently of black-hole physics [28]. Closely related singular-value diagnostics appear in pseudospectrum analyses of non-self-adjoint QNM operators, where contours of resolvent norms or smallest singular values measure spectral sensitivity [26, 27]. The present work uses this existing viewpoint as a practical KS computation and validation workflow: locate residual minima, refine them with the spectral operator, and check them against branch convergence and a Leaver-style continued-fraction solver.

2 Methods

2.1 Kazakov–Solodukhin perturbation problem

Kazakov and Solodukhin obtained a spherically symmetric quantum deformation of the Schwarzschild solution by reducing the four-dimensional Einstein theory to an effective two-dimensional dilaton-gravity problem and choosing a renormalizable dilaton potential [14]. We use the areal-radius form subsequently employed in the KS QNM analysis of Konoplya [16]:

$$\begin{aligned} ds^2 &= -f_a(r)dt^2 + f_a(r)^{-1}dr^2 + r^2 d\Omega^2, \\ f_a(r) &= \frac{\sqrt{r^2-a^2}}{r} - \frac{2M}{r}, \end{aligned} \tag{1}$$

Here r is the areal radius, $M > 0$ is the mass parameter, and $a \geq 0$ is a length scale. No radial-coordinate transformation relative to the convention of Ref. [16] is made. In that convention the present numerical work treats a directly as the deformation parameter; no otherwise unused renormalized coupling is introduced. Reality of the metric coefficients restricts the coordinate patch to $r \geq a$. The outer horizon follows from $f_a(r_h) = 0$:

$$\sqrt{r_h^2 - a^2} = 2M, \qquad r_h = \sqrt{4M^2 + a^2}. \tag{2}$$

Thus $r_h > a$ for every finite a/M, and the value $a/M = 1$ used below is not an extremal or degenerate-horizon limit; it is the upper endpoint of the deformation interval sampled in this study. Finally, $\sqrt{r^2 - a^2}/r \to 1$ as $a \to 0$, so $f_a(r) \to f_0(r) = 1 - 2M/r$ and Eq. (1) reduces directly to Schwarzschild in standard areal coordinates.

The first perturbation sector is a minimally coupled massless test scalar. Starting from

$$\Box\Phi = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\, g^{\mu\nu}\partial_\nu\Phi\right) = 0 \tag{3}$$

and separating variables according to

$$\Phi(t, r, \theta, \phi) = e^{-i\omega t} Y_{\ell m}(\theta, \phi)\frac{\Psi(r)}{r}, \tag{4}$$

with $\nabla^2_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$, gives

$$\frac{1}{r^2}\frac{d}{dr}\left[r^2 f_a(r)\frac{d}{dr}\left(\frac{\Psi}{r}\right)\right] + \left[\frac{\omega^2}{f_a(r)} - \frac{\ell(\ell+1)}{r^2}\right]\frac{\Psi}{r} = 0. \tag{5}$$

Introducing the tortoise coordinate $dr_*/dr = f_a^{-1}$ reduces this equation to the Schrödinger form

$$\frac{d^2\Psi}{dr_*^2} + \left[\omega^2 - V_{0\ell}(r; a)\right]\Psi = 0, \tag{6}$$

where the scalar effective potential is therefore

$$V_{0\ell}(r;a) = f_a(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{1}{r}\frac{\mathrm{d}f_a}{\mathrm{d}r}\right], \tag{7}$$

in agreement with the minimally coupled scalar equation used in Ref. [16]. QNM boundary conditions select solutions that are purely ingoing at the horizon and purely outgoing at spatial infinity. These conditions make ω complex and the radial problem non-self-adjoint.

The odd-parity sector requires an assumption beyond the spherical KS background. The original KS construction explicitly writes the full metric as a nonperturbative spherically symmetric part plus small propagating nonspherical modes and treats the latter classically at leading order [14]. Motivated by that separation, we represent the quantum-corrected spherical background through $G_{\mu\nu} = 8\pi T^{\mathrm{eff}}_{\mu\nu}$ and set the *gauge-invariant odd-parity current* of the effective source to zero. This is an additional inverse-Cowling (frozen-source) closure, not a constitutive law derived from the spherical KS effective action. It retains the KS spherical quantum backreaction but sets the odd part of $\delta T^{\mathrm{eff}}_{\mu\nu}$ to zero and therefore neglects independent nonspherical quantum-source excitations. The modes below are conditional on this explicitly defined model.

To display the gauge invariance, write the background as $g_{ab}(y)\mathrm{d}y^a\mathrm{d}y^b + r^2(y)\gamma_{AB}\mathrm{d}z^A\mathrm{d}z^B$, where $a,b \in \{t,r\}$, and expand an odd perturbation as

$$\delta g_{aB} = h_a X^{\ell m}_B, \qquad \delta g_{AB} = h_2 X^{\ell m}_{AB}. \tag{8}$$

Under an odd infinitesimal diffeomorphism generated by $\xi_A = \Lambda X^{\ell m}_A$,

$$h_a \mapsto h_a - D_a\Lambda + \frac{2D_a r}{r}\Lambda, \qquad h_2 \mapsto h_2 - 2\Lambda. \tag{9}$$

With our harmonic normalization $h_2 = 2h$, the one-form

$$k_a = h_a - \frac{1}{2}D_a h_2 + \frac{D_a r}{r}h_2 \tag{10}$$

is therefore gauge invariant. This is Eq. (18) of Gundlach and Martínez-García [3], written in the Gerlach–Sengupta conventions of Ref. [2]. We choose the two-dimensional volume form so that $\epsilon^{tr} = +1$ in the static (t,r) chart and define, as in Eq. (68) of Ref. [3],

$$\Pi = \epsilon^{ab} D_b\left(\frac{k_a}{r^2}\right). \tag{11}$$

Writing the odd stress perturbation as $\delta T_{aB} = \delta t^{\mathrm{odd}}_a X_B$, its gauge-invariant source one-form is $L_a = \delta t^{\mathrm{odd}}_a - Q_{\mathrm{bg}} h_a$, where Q_{bg} denotes the coefficient called Q for the angular background stress tensor in the notation of Ref. [3]; this is their Eq. (19). Because it is constructed from the gauge-invariant matter combination, imposing $L_a = 0$ does not fix a coordinate gauge. The general sourced master equation, Eq. (69) of Ref. [3], is

$$D_a\left[\frac{1}{r^2}D^a(r^4\Pi)\right] - (\ell-1)(\ell+2)\Pi = -16\pi\epsilon^{ab}D_b L_a. \tag{12}$$

The associated conservation law is $D_a(r^2L^a) = (\ell-1)(\ell+2)L$, Eq. (62) of the same reference. Our inverse-Cowling closure sets both odd source amplitudes $L_a = L = 0$, so the conservation equation and the right-hand side of Eq. (12) vanish identically. Equivalently, Karlovini's Eqs. (38), (39), and (46) give $(D^a D_a - U)\psi = \kappa r \epsilon^{ab} D_a (r^{-2} J_b)$, with the same source-free reduction [4].

For the static metric used here, the normalization $\Psi_{\rm ax} = r^3 \Pi$ (up to an irrelevant constant) and $\mathrm{d}r_*/\mathrm{d}r = f_a^{-1}$ remove the first radial derivative from the source-free equation. After Fourier decomposition with $e^{-\mathrm{i}\omega t}$, it becomes

$$\frac{\mathrm{d}^2\Psi_{\rm ax}}{\mathrm{d}r_*^2} + \left[\omega^2 - V_{\mathrm{ax},\ell}(r;a)\right]\Psi_{\rm ax} = 0, \tag{13}$$

with the general anisotropic-source potential

$$V_{\mathrm{ax},\ell} = f_a \left[\frac{\ell(\ell+1)}{r^2} - \frac{6m(r)}{r^3} + 4\pi\big(\rho - p_r\big)\right], \qquad f_a = 1 - \frac{2m(r)}{r}. \tag{14}$$

This form is also obtained in the inverse-Cowling treatment of general static spherical backgrounds [Eqs. (14)–(17) of Ref. [5]].

For the KS metric, $g_{tt}g_{rr} = -1$. The background Einstein identities give

$$m(r) = \frac{r}{2}(1 - f_a) = M + \frac{r - \sqrt{r^2 - a^2}}{2}, \quad 4\pi\rho = \frac{m'}{r^2}, \quad p_r = -\rho. \tag{15}$$

Substitution into Eq. (14) yields the gauge-invariant inverse-Cowling KS axial potential used in all calculations:

$$V_{\mathrm{ax},\ell}(r;a) = f_a(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{2(f_a - 1)}{r^2} - \frac{f_a'}{r}\right]. \tag{16}$$

At $a = 0$, $m = M$, $\rho = p_r = 0$, and Eq. (16) becomes the Schwarzschild Regge–Wheeler potential [1]. For $a > 0$, it differs from the earlier lapse-substitution ansatz by

$$\Delta V_{\rm ax} = \frac{f_a}{r^3}\left[2\left(\sqrt{r^2 - a^2} - r\right) - \frac{a^2}{\sqrt{r^2 - a^2}}\right], \tag{17}$$

so it is not a relabeling of the earlier lapse-substitution ansatz. The same effective-source potential is also used in the public high-precision Chebyshev calculation of Batic, Dutykh, and Sukaiti [6]; Section 3 compares overlapping modes directly. Our distinct contribution is the audited low-lying workflow: residual and backward-error diagnostics, continuation and confidence labels, a collocation-independent continued-fraction check, and local pseudospectral sensitivity tests. Gauge invariance removes coordinate artifacts from the metric master variable; it does not make the inverse-Cowling closure unique. A nonspherical quantum-source action could instead generate a nonzero right-hand side or additional couplings in Eq. (12).

2.2 Numerical method

The implementation now contains two deliberately separated paths. The first is a baseline diagnostic: the scalar wave equation is evolved in time, a ringdown window is fitted, and a matrix-pencil frequency is extracted from the waveform. The second is the application-specific contribution of this paper: a direct Chebyshev pseudospectral discretization of the KS perturbation equation, followed by construction and audit of the Hermitian residual $R_N(\omega) = P_N(\omega)^\dagger P_N(\omega)$. The matrix-pencil operator is not used as the final QNM operator.

2.2.1 Baseline time-domain diagnostic

The baseline evolves the radial wave equation in tortoise coordinate $x = r_*$,

$$\frac{\partial^2\Psi}{\partial t^2} - \frac{\partial^2\Psi}{\partial x^2} + V_{0\ell}(r(x); a)\Psi = 0. \tag{18}$$

Equation (18) is used only as a baseline diagnostic. The computational domain is $x/M \in [-250, 500]$, with $\Delta x = 0.1M$, $\Delta t = 0.05M$, $t_{\max} = 320M$, and observation point $x_{\text{obs}} = 20M$. A Gaussian pulse $\Psi(0, x) = \exp[-(x/4M)^2]$ is evolved with first-order outgoing boundary conditions. The interval $60M \le t \le 160M$ is fitted by

$$\Psi_{\text{fit}}(t) = Ae^{-\alpha(t-t_0)}\cos(\omega_R t + \phi) + C_0, \qquad \omega_{\text{fit}} = \omega_R - \mathrm{i}\alpha.$$

A rank-four matrix-pencil reduction is also applied to the fitted waveform as a diagnostic frequency extractor. This path is useful because it gives a physically intuitive check on the deformation trend, but it is no longer the operator used in the residual calculation.

2.2.2 Chebyshev spectral discretization

The direct spectral method compactifies the horizon-to-infinity domain with

$$s = 1 - \frac{r_h}{r}, \qquad x = 2s - 1, \qquad x \in [-1, 1],$$

where $r_h = \sqrt{4M^2 + a^2}$. Chebyshev-Gauss collocation points

$$x_j = \cos\left(\frac{\pi(j + 1/2)}{N}\right), \qquad j = 0, \dots, N - 1,$$

are mapped to $s_j = (x_j + 1)/2$, excluding the two singular endpoints. Barycentric differentiation matrices D and D^2 approximate ∂_s and ∂_s^2.

The QNM boundary behavior is imposed analytically by factoring

$$\Psi(r) = e^{\mathrm{i}\omega r}\, r^{2\mathrm{i}M\omega}\, s^{-\mathrm{i}\omega/f_a'(r_h)}\, u(s). \tag{19}$$

The factorization in Eq. (19) enforces outgoing behavior at infinity and ingoing behavior at the horizon, while $u(s)$ is regular on the collocation grid. Substitution into Eq. (6) gives a quadratic polynomial eigenvalue problem

$$P_N(\omega)\boldsymbol{u} \equiv \left(A_0 + \omega A_1 + \omega^2 A_2\right)\boldsymbol{u} = 0. \tag{20}$$

The matrices A_0, A_1, A_2 are built directly from $f_a(r)$, $f_a'(r)$, the selected potential $V_{0\ell}$ or $V_{\text{ax},\ell}$, and the chain-rule factors relating r and s.

2.2.3 Spectral residual workflow

The workflow object of the paper is the residual map

$$\omega \mapsto \sigma_{\min}(P_N(\omega)),$$

where $P_N(\omega)$ is the finite-dimensional Chebyshev discretization of the factored QNM equation. The workflow is:

Input.

- a perturbation equation and QNM boundary conditions, such as Eq. (6);
- a spectral size N, which fixes the Chebyshev grid and matrix dimension;
- a search region in complex ω, chosen from Schwarzschild references, continuation in a/M, or a physically damped frequency window.

Procedure.

1. Build the polynomial spectral operator $P_N(\omega) = A_0 + \omega A_1 + \omega^2 A_2$.
2. Construct the Hermitian residual $R_N(\omega) = P_N(\omega)^\dagger P_N(\omega)$.
3. Compute $\sigma_{\min}(P_N(\omega))$, equivalently the square root of the smallest eigenvalue of $R_N(\omega)$ when roundoff is controlled.
4. Search for local minima of $\sigma_{\min}(P_N(\omega))$ in the selected complex-frequency region and use generalized-eigenvalue roots as initialization when available.
5. Validate candidate frequencies by convergence in N, branch continuation in a/M, residual size, and comparison with the Leaver-style continued-fraction solver.

Output. The accepted outputs are QNM frequencies $M\omega_{\ell n}(a/M)$, together with residual norms, branch labels, scale-aware backward-error diagnostics, and validation status. This acceptance layer, rather than the raw generalized-eigenvalue list or the absolute accuracy of one Schwarzschild benchmark, is the methodological contribution of the workflow.

2.2.4 Generalized eigenproblem and residual

Equation (20) is linearized as

$$\begin{pmatrix} 0 & I \\ -A_0 & -A_1 \end{pmatrix} \begin{pmatrix} \boldsymbol{u} \\ \omega \boldsymbol{u} \end{pmatrix} = \omega \begin{pmatrix} I & 0 \\ 0 & A_2 \end{pmatrix} \begin{pmatrix} \boldsymbol{u} \\ \omega \boldsymbol{u} \end{pmatrix}, \tag{21}$$

The generalized eigenproblem in Eq. (21) is solved classically with `scipy.linalg.eig`. These generalized-eigenvalue roots supply efficient initial candidates for the residual search. The fundamental scalar $\ell = 2$ mode is selected by proximity to the Schwarzschild reference frequency; overtones are tracked by continuity, branch diagnostics, and validation status rather than by proximity alone.

2.2.5 Candidate filtering, refinement, and branch assignment

The operational selection rules are fixed in the implementation rather than chosen interactively. Before candidate generation, each row of the pair of linearized matrices is divided by the larger of its two row infinity norms; the same operation is then applied columnwise after row scaling. Scaling factors are clipped to $[10^{-100}, 10^{100}]$, and the right generalized eigenvectors are mapped back to the unscaled coordinates. Infinite and non-finite roots are removed. The physical search window is $0.05 < \mathrm{Re}\,\omega < 2$ and $-3 < \mathrm{Im}\,\omega < -0.02$. Roots separated by at most 10^{-7} are treated as one numerical cluster, retaining the representative with the smaller raw-polynomial singular-value residual. At most the 20 candidates closest to the current branch target enter the detailed scoring stage.

For a previously tracked branch, a candidate must lie within an absolute complex-frequency distance 0.20 of the preceding target. Competing matches minimize

$$S = d_{\mathrm{target}} + 0.45 d_{\mathrm{cont}} + 0.35(1 - |\langle u_{j-1}, u_j \rangle|) + 10^{-3} \min(\sigma_j/\sigma_{\mathrm{best}}, 100),$$

where, for the shortlisted candidate set C,

$$d_{\text{target}} = \frac{|\omega_j - \omega_{\text{target}}|}{\max(|\omega_{\text{target}}|, 0.1)}, \qquad d_{\text{cont}} = \frac{|\omega_j - \omega_{j-1}|}{\max(|\omega_{j-1}|, 0.1)}, \qquad \sigma_{\text{best}} = \min_{k\in C} \sigma_k.$$

Here $\sigma_j = \sigma_{\min}[P_N(\omega_j)]$ is evaluated on the raw, unequilibrated polynomial matrices at the unrefined generalized-eigenvalue candidate. Candidate frequencies and right eigenvectors are generated from the row- and column-equilibrated linearized pencil; before overlap evaluation, the right eigenvectors are mapped back by the column-scaling matrix as $x = D_r y$. The extracted mode shapes are normalized, and the previous shape is barycentrically interpolated to the new grid. The absolute value in $|\langle u_{j-1}, u_j\rangle|$ removes the arbitrary complex phase of either eigenvector. The selected raw root is refined within a rectangular radius 0.02 in each complex-frequency component by L-BFGS-B minimization of $\sigma^2_{\min}[P_N(\omega)]$, with at most 500 iterations, `ftol`$= 10^{-18}$, and `gtol`$= 10^{-12}$.

The catalogue labels $n = 0, 1, 2$ begin from the stated Schwarzschild seeds; after each deformation value, the corresponding continued-fraction roots become the three targets for the next value, and already assigned roots are excluded within 10^{-7}. A catalogue row passes the automated spectral gate only when its raw-polynomial backward error is below 10^{-8} and its spectral–continued-fraction relative difference is below 10^{-4}. The confidence tier is nevertheless retained: passing this numerical gate does not promote $n = 2$ beyond exploratory status.

In pseudocode, the implemented path is: (1) equilibrate and solve the quadratic pencil; (2) remove non-finite roots and roots outside the damped window; (3) cluster at 10^{-7}; (4) enforce the 0.20 continuation gate and score competing matches; (5) refine the selected Chebyshev root by bounded singular-value minimization; (6) apply the backward-error and spectral–continued-fraction gates; and (7) propagate the validated continued-fraction root as the next deformation target. Thus the continued-fraction calculation is collocation independent, although it shares the perturbation equation and endpoint factorization.

The residual framework is unchanged. The direct spectral residual is

$$R_N(\omega) = P_N(\omega)^\dagger P_N(\omega). \tag{22}$$

Equation (22) is Hermitian and positive semidefinite by construction, up to numerical roundoff. The residual norm reported below is the smallest singular value of $P_N(\omega)$, and the residual-minimized frequency is obtained by locally minimizing this singular value over complex ω. In the tables below this residual is used together with the scale-aware polynomial backward error; neither is interpreted as a precision guarantee without spectral-size behavior and continued-fraction validation.

2.2.6 Leaver-style validation

The Chebyshev spectral operator remains the primary operator path, but a Leaver-style validation module is added. It uses the same compact coordinate $s = 1 - r_h/r$ and the same QNM endpoint factorization, but does not use Chebyshev collocation, matrix-pencil data, or the residual observable. Instead it writes

$$\Psi(r) = e^{\mathrm{i}\omega r} r^{2\mathrm{i}M\omega} s^{-\mathrm{i}\omega/f'(r_h)} u(s),$$

and expands the regular factor as a Frobenius series,

$$u(s) = \sum_{n=0}^{\infty} a_n s^n.$$

Substitution into the factored radial equation gives

$$B_2(s)u'' + B_1(s)u' + B_0(s)u = 0,$$

where the $B_j(s)$ are generated by formal power-series algebra from the KS metric functions. The Taylor-truncated Frobenius equations define a multi-term recurrence for the coefficients a_n. Gaussian elimination reduces this recurrence to the three-term form

$$\alpha_n a_{n+1} + \beta_n a_n + \gamma_n a_{n-1} = 0,$$

and the minimal-solution condition is imposed through Leaver's continued fraction,

$$\beta_0 - \cfrac{\alpha_0\gamma_1}{\beta_1 - \cfrac{\alpha_1\gamma_2}{\beta_2 - \cdots}} = 0. \tag{23}$$

The reported calculation expands each B_j through Taylor order 96 and evaluates the reduced recurrence to continued-fraction depth 240. Roots of Eq. (23) are obtained in IEEE-754 double precision by applying the MINPACK hybrid solver exposed by `scipy.optimize.root` to the real and imaginary residual components. The root tolerance is 10^{-11}, the maximum number of residual-function evaluations is 1000, and the accepted residual is required to be below 10^{-7} for the scalar validation calculation. The initial guesses are the branch-tracked $N = 32$ Chebyshev candidates. Branch identity is maintained by continuation over $a/M = 0, 0.2, 0.5, 1$ and by keeping the fundamental and first-overtone targets distinct. The catalogue uses the same settings, with a relaxed 10^{-4} residual acceptance threshold for the most delicate second-overtone recurrence reductions. This validation layer is independent of Chebyshev collocation, matrix-pencil data, and residual minimization, while sharing the same perturbation equation and endpoint factorization.

3 Results

3.1 Schwarzschild validation

As an external benchmark we use the spin-zero Schwarzschild fundamental with $\ell = 2$, overtone index $n = 0$, mass normalization $M = 1$, time dependence $e^{-\mathrm{i}\omega t}$, and therefore $\operatorname{Im}\omega < 0$ for a damped mode. Cavalcante and da Cunha report this value in Table I on page 6 from an isomonodromic Painlevé calculation and reproduce it in the adjacent column with a continued-fraction method [29]:

$$M\omega_{\mathrm{ref}} = 0.483643872211 - 0.096758775978\,\mathrm{i}.$$

Throughout the numerical comparisons we use

$$\Delta_{\mathrm{rel}}(\omega_1, \omega_2) = \frac{|\omega_1 - \omega_2|}{|\omega_2|}.$$

At $N = 96$, the direct spectral calculation gives

$$M\omega_{\mathrm{spec}} = 0.483643872189 - 0.096758775961\,\mathrm{i},$$

with relative difference 5.60×10^{-11} from that external value. This is substantially sharper than the baseline time-domain fit, $0.483642480 - 0.096758919\,\mathrm{i}$, and the waveform matrix-pencil diagnostic, $0.483653530 - 0.096763421\,\mathrm{i}$.

The Leaver-style solver gives

$$M\omega_{\mathrm{Leaver}} = 0.483643872211 - 0.096758775978\,\mathrm{i}$$

for the Schwarzschild fundamental mode, with relative difference 8.24×10^{-13} from the same external value. The point of this benchmark is not to claim a more accurate Schwarzschild solver than established high-precision methods. It verifies that two discretizations used by the audit workflow identify the same low-lying branch and reproduce an independently published value under explicitly matched conventions.

The convergence of the direct spectral branches with Chebyshev size is shown in Figure 1. The reference in every curve is the collocation-independent order-128, depth-320 continued-fraction root, not another Chebyshev result. The fundamental reaches a double-precision plateau, whereas the first overtone is closest to the continued-fraction root around $N = 32$–48 and then deteriorates. This nonmonotonic behavior is why the first overtone is described as cross-validated at $N = 32$, not as a high-N spectral result.

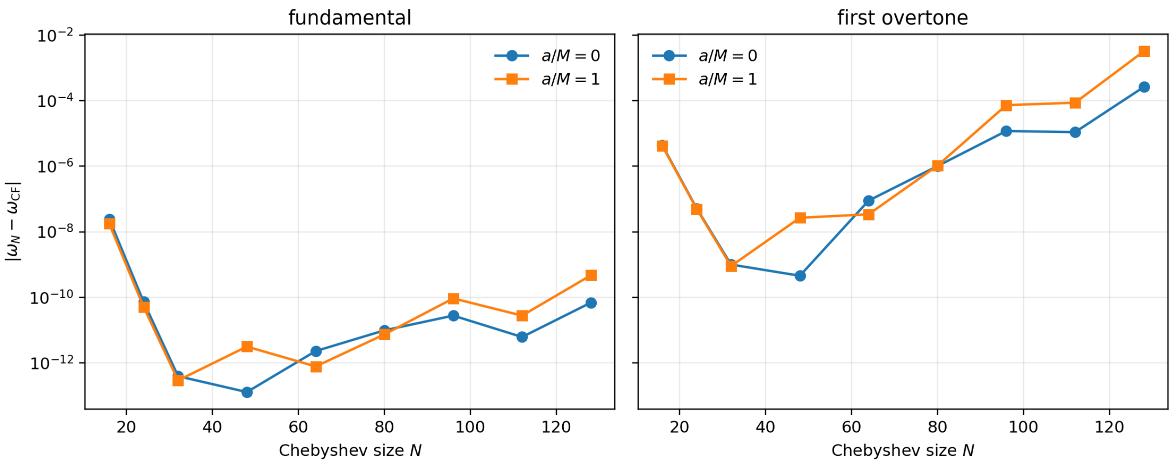

Figure 1 Chebyshev error relative to the order-128, depth-320 continued-fraction root for scalar $\ell = 2$ fundamentals (left) and first overtones (right), including $N = 128$. The overtone panel displays the high-N deterioration directly.

As a separate upper-grid spot check, the scalar $\ell = 2, n = 0$ branch was recomputed at $N = 128$ for $a/M = 0$ and $a/M = 1$. The $N = 128$ values differ from the reported $N = 96$ values by 6.95×10^{-11} and 4.64×10^{-10}, respectively, and are tabulated by `scripts/check_n128_spot.py`. These differences do not change any reported percentage shift or catalogue trend. Because the $N = 96 \longrightarrow 128$ movement is part of a double-precision high-N plateau rather than a monotonic gain of significant digits, the paper keeps $N = 96$ plus Leaver validation as the reference. The high-N plateau is consistent with roundoff amplification in the differentiation matrices and linearized pencil. Accuracy is therefore assessed using spectral-size behavior, polynomial backward error, and the collocation-independent continued-fraction result; the plateau should not be interpreted as unresolved physical drift.

3.2 Leaver continued-fraction comparison

Table 1 summarizes the Leaver-style continued-fraction validation against the targeted Chebyshev spectral rows used for the scalar comparison.

a/M	mode	$\mathrm{Re}(M\omega_{\mathrm{Leaver}})$	$\mathrm{Im}(M\omega_{\mathrm{Leaver}})$	$\Delta_{\mathrm{rel}}(\mathrm{spec, Leaver})$
0.0	fundamental	0.48364387	-0.09675878	7.914×10^{-13}
0.0	first overtone	0.46385058	-0.29560394	1.814×10^{-9}
0.2	fundamental	0.48203369	-0.09665386	6.084×10^{-13}
0.2	first overtone	0.46216559	-0.29531130	1.788×10^{-9}
0.5	fundamental	0.47388693	-0.09610918	6.219×10^{-13}
0.5	first overtone	0.45364275	-0.29379018	1.808×10^{-9}
1.0	fundamental	0.44836241	-0.09424818	6.341×10^{-13}
1.0	first overtone	0.42697149	-0.28857291	1.747×10^{-9}

Table 1 Leaver-style continued-fraction validation against targeted Chebyshev spectral results at $N = 32$. Frequency columns are rounded for display; the discrepancy column is computed from full-precision internal values. The Leaver solver avoids Chebyshev collocation, matrix-pencil data, and residual minimization, while sharing the same perturbation equation and endpoint factorization. The stricter scalar comparison shown here fails if the relative difference exceeds 10^{-6}; the looser general catalogue gate of 10^{-4} exists only to accommodate the exploratory $n = 2$ tier. The largest difference in this table is 1.814×10^{-9}.

Table 2 isolates continued-fraction truncation from the spectral comparison. The fundamental modes stabilize by depth 90 at the shown precision. The deformed first overtone is more sensitive, but its depth-240 and depth-320 values differ by only 2.05×10^{-12}. This is consistent with treating scalar $n = 1$ as a cross-validated low-lying first overtone at $N = 32$ while retaining a stricter claim for the fundamentals.

case	CF depth	$\mathrm{Re}(M\omega)$	$\mathrm{Im}(M\omega)$	$\lvert\omega - \omega_{320}\rvert$
Schwarzschild $n = 0$	60	0.483643872211	-0.096758775978	2.68×10^{-13}
	90	0.483643872211	-0.096758775978	4.89×10^{-16}
	120	0.483643872211	-0.096758775978	0
	240	0.483643872211	-0.096758775978	0
	320	0.483643872211	-0.096758775978	0
KS $a/M = 1, n = 0$	60	0.448362409002	-0.094248179159	1.76×10^{-13}
	90	0.448362409002	-0.094248179159	1.35×10^{-15}
	120	0.448362409002	-0.094248179159	3.93×10^{-16}
	240	0.448362409002	-0.094248179159	5.55×10^{-17}
	320	0.448362409002	-0.094248179159	0
KS $a/M = 1, n = 1$	60	0.426971489725	-0.288572913307	2.09×10^{-10}
	90	0.426971489574	-0.288572913453	1.12×10^{-12}
	120	0.426971489576	-0.288572913449	2.84×10^{-12}
	240	0.426971489574	-0.288572913453	2.05×10^{-12}
	320	0.426971489574	-0.288572913452	0

Table 2 Continued-fraction depth check at fixed Taylor order 96. Each solve uses the branch-tracked $N = 32$ Chebyshev value only as an initial guess; the last column compares with the depth-320 root. The table is generated by `scripts/analyze_leaver_convergence.py`.

Taylor-order convergence was tested independently at fixed continued-fraction depth 320. Table 3 reports the complex-frequency difference from order 128. At the production order 96 the largest difference among these cases is 4.32×10^{-12}.

case	64	80	96	112	128
scalar $\ell = 2, n = 0, a/M = 1$	2.90×10^{-13}	1.60×10^{-14}	2.50×10^{-15}	0	0
scalar $\ell = 2, n = 1, a/M = 1$	1.20×10^{-9}	4.32×10^{-12}	4.32×10^{-12}	3.30×10^{-13}	0
axial $\ell = 2, n = 0, a/M = 1$	3.09×10^{-12}	1.67×10^{-13}	6.13×10^{-14}	3.40×10^{-15}	0

Table 3 Absolute difference $|\omega_p - \omega_{128}|$ under independent variation of the Frobenius Taylor order p, with continued-fraction depth 320 fixed. Generated by `scripts/analyze_solver_convergence.py`.

3.3 Gauge-invariant odd-parity gravitational sector

The same Chebyshev and continued-fraction infrastructure is applied to the gauge-invariant inverse-Cowling axial equation for $\ell = 2, 3, 4$ and low-lying indices $n = 0, 1, 2$. Together with the scalar rows, the working catalogue contains 72 entries over $a/M = 0, 0.2, 0.5, 1.0$. The axial entries are conditional inverse-Cowling modes of the explicitly stated closure, rather than modes of the former lapse-substitution proxy. At $a = 0$, the effective source vanishes and the model reduces exactly to the Schwarzschild Regge–Wheeler equation. Table 4 lists the resulting continued-fraction frequencies. External high-precision comparisons are reported separately below.

ℓ	n	$\mathrm{Re}(M\omega_{\mathrm{Leaver}})$	$\mathrm{Im}(M\omega_{\mathrm{Leaver}})$
2	0	0.37367168	-0.08896232
2	1	0.34671100	-0.27391488
2	2	0.30105346	-0.47827698
3	0	0.59944329	-0.09270305
3	1	0.58264380	-0.28129811
3	2	0.55168490	-0.47909275
4	0	0.80917838	-0.09416396
4	1	0.79663153	-0.28433435
4	2	0.77270953	-0.47990818

Table 4 Schwarzschild Regge–Wheeler frequencies obtained by the continued-fraction solver. The former relative-error column has been removed because the transcribed literature values had uneven precision and therefore did not provide a uniform solver-accuracy metric. Agreement with the named high-precision external calculation is reported separately below.

Representative gauge-invariant inverse-Cowling axial $\ell = 2$ trajectories over the deformation grid are shown in Figure 2.

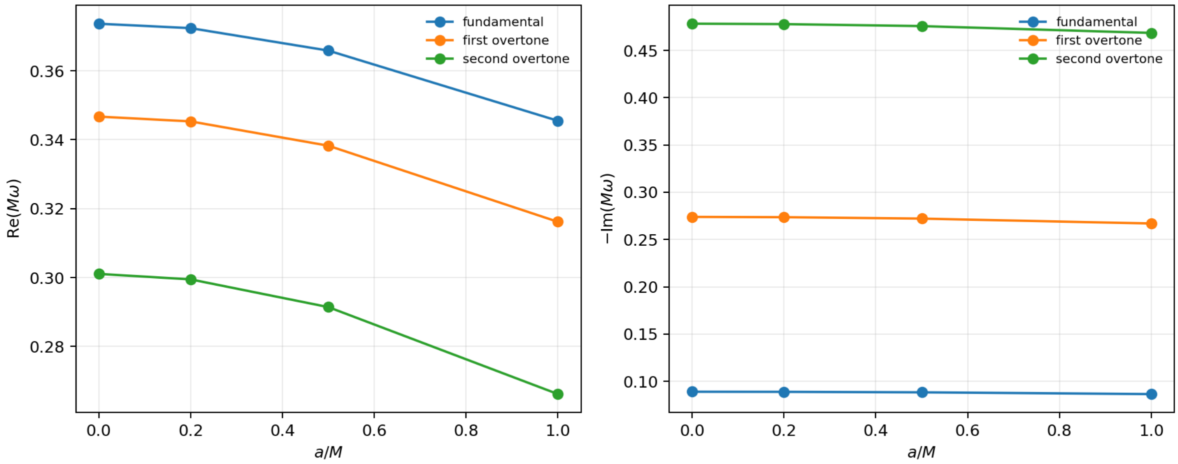

Figure 2 Representative $\ell = 2$ conditional odd-parity mode trajectories of the gauge-invariant inverse-Cowling effective-source model versus a/M. They are not predictions of an unspecified quantum nonspherical completion.

Figure 3 compares the inverse-Cowling potential with the former lapse-substitution ansatz. At $a/M = 1$, the maximum absolute correction is 5.77% of the new barrier height; over the region where the new potential exceeds 10^{-3} of its peak, the maximum pointwise fractional correction is 15.1%. These definitions avoid a spurious relative divergence at the horizon and infinity, where both potentials vanish.

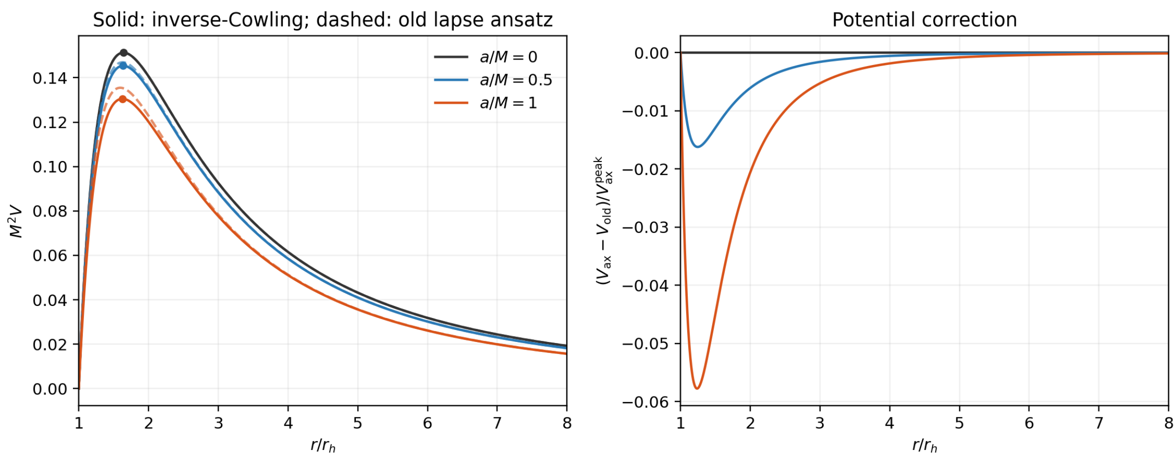

Figure 3 Left: inverse-Cowling axial potential (solid) and the former lapse-substitution ansatz (dashed) for $\ell = 2$; circles mark the new potential peaks and the vertical line marks $r = r_h$. Right: their difference normalized by the new barrier height.

The same potential and mass normalization are used by the public 2026 high-precision Chebyshev calculation of Batic, Dutykh, and Sukaiti [6]. We compared all 18 overlapping $a/M = 0.5, 1$, $\ell = 2, 3, 4$, $n = 0, 1, 2$ entries. The largest relative difference is 1.98×10^{-13} for fundamentals, 3.80×10^{-10} for first overtones, and 1.62×10^{-5} for second overtones. The last occurs at $a/M = 1, \ell = 2, n = 2$ and reinforces the exploratory status of that tier. The comparison, parameter conversion check, public repository commit, and local source commit are recorded by `scripts/audit_axial_model.py`.

As a stability diagnostic, the potential was minimized numerically outside the horizon for every sampled a/M and $\ell = 2, 3, 4$. It remained non-negative and approached zero only at the two boundaries. No candidate with $\mathrm{Im}\,\omega > 0$ survived a raw-spectrum comparison at $N = 32, 48, 64$. Positivity supplies the usual sufficient mode-stability argument for this source-free one-dimensional problem. For a hypothetical mode with $\mathrm{Im}\,\omega > 0$, the QNM boundary conditions make the radial solution decay at both asymptotic ends; multiplying the source-free master equation by Ψ^*_{ax}, integrating, and using the standard positive-energy identity excludes a growing mode when $V_{\mathrm{ax}} \geq 0$ [5]. This statement applies only to the inverse-Cowling model and does not establish stability under other effective-source closures; the finite root search is an additional numerical check.

Across the low-lying working catalogue, the worst spectral–Leaver relative difference is 1.202×10^{-5}, occurring for the axial $\ell = 2$, $n = 2$ branch at $a/M = 0.5$. The conditional $n = 0$ axial fundamentals are numerically the most robust axial tier, while axial $n = 1$ and $n = 2$ entries remain conditional inverse-Cowling odd-parity modes; $n = 2$ entries are exploratory branch diagnostics rather than equal-confidence physical predictions. Higher overtones are outside the scope.

3.4 Catalogue-level deformation trends

The catalogue now supports a simple physics-facing analysis: each branch is compared against its own Schwarzschild value at $a/M = 0$. The real frequency and damping magnitude decrease successively across the four sampled deformation values for every scalar branch; the gauge-invariant inverse-Cowling axial branches show the same sampled-grid trend. The scalar result remains the least assumption-dependent physical statement.

At the upper endpoint of the sampled interval, $a/M = 1$, the scalar $\ell = 2$ fundamental shifts are -7.29% in $\mathrm{Re}(M\omega)$, -2.59% in the damping magnitude $-\mathrm{Im}(M\omega)$, and 7.17% in the complex-plane displacement $|\Delta\omega|/|\omega(a = 0)|$. These are distinct quantities and are not referred to interchangeably below. Among the scalar fundamentals, the $\ell = 4$ branch has the largest complex-plane displacement, 7.23%, while its real-part shift is -7.27%. The scalar $\ell = 2$ second overtone has a larger real-part shift, -9.16%, but is an exploratory branch diagnostic. The $\ell = 2$ trends are plotted in Figure 4.

Within the closure-dependent axial fundamentals, the largest complex-plane displacement is 7.37% for $\ell = 2$, with shifts of -7.54% in the real part and -2.80% in damping magnitude. This value is reported separately because it depends on the frozen-source assumption, whereas the scalar comparison above does not.

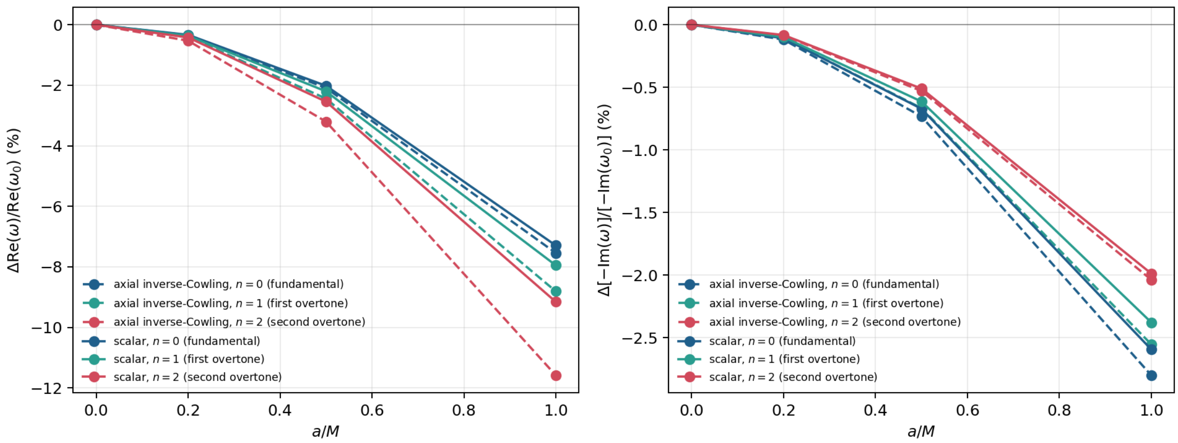

Figure 4 Schwarzschild-relative trends over the sampled interval for the $\ell = 2$ branches. Solid curves show scalar modes and dashed curves show the gauge-invariant inverse-Cowling axial modes. The latter are conditional on the closure stated in Section 2.1.

The small increase of the scalar fundamental displacement from $\ell = 2$ to $\ell = 4$ can be checked against the effective-potential barrier rather than explained speculatively. Table 5 reports the peak audit at $a/M = 1$. The leading WKB proxy $\mathrm{Re}\,\omega \sim \sqrt{V_{\mathrm{peak}}}$ changes by -7.01%, -7.13%, and -7.18% for $\ell = 2, 3, 4$, respectively, mirroring the modest multipole dependence of the computed fundamentals. The curvature change also grows slightly with ℓ, consistent with a multipole-dependent damping response. This supports a cautious potential-barrier interpretation; it does not indicate a qualitatively new $\ell = 4$ mechanism.

ℓ	r_{peak}/M	ΔV_{peak}	$\Delta\sqrt{V_{\mathrm{peak}}}$	$\Delta\lvert V''_{*,\mathrm{peak}}\rvert$
2	3.25198574	−13.5350%	−7.0134%	−17.7698%
3	3.28015848	−13.7535%	−7.1310%	−18.1334%
4	3.29264164	−13.8440%	−7.1797%	−18.2801%

Table 5 Scalar effective-potential peak changes at $a/M = 1$, relative to the Schwarzschild peak at the same ℓ. Primes denote tortoise-coordinate derivatives. These diagnostics are generated by `scripts/analyze_scalar_potential_peaks.py` and provide only WKB intuition; the QNM frequencies are obtained from the Chebyshev–Leaver workflow.

3.5 Spectroscopy diagnostics

At fixed mass scale, the catalogue trend admits a spectroscopic interpretation. The negative shift in $\mathrm{Re}(M\omega)$ lowers the test-scalar oscillation frequency, while the negative shift in $-\mathrm{Im}(M\omega)$ lowers the absolute damping rate and therefore lengthens the damping time. We define the quality factor consistently as

$$Q = \frac{\mathrm{Re}\,\omega}{2[-\mathrm{Im}\,\omega]}.$$

For the scalar $\ell = 2$ fundamental branch, the damping magnitude decreases by 2.59%, corresponding to an approximate 2.7% increase in damping time. Because the oscillation frequency decreases by a larger fraction, Q decreases by 4.83%: the scalar perturbation decays slightly more slowly, but it completes fewer oscillations per damping time.

Dimensionless mode ratios sharpen this interpretation because they are less directly absorbable into a simple mass rescaling than individual frequencies. For the scalar $\ell = 2$ branch at $a/M = 1$, the complex ratio ω_0/ω_1 changes from $0.836060 + 0.324207\,\mathrm{i}$ to $0.823241 + 0.335659\,\mathrm{i}$, a 1.92% shift in the complex-ratio plane. The corresponding ω_0/ω_2 ratio changes from $0.579814 + 0.460140\,\mathrm{i}$ to $0.553881 + 0.464905\,\mathrm{i}$, a 3.56% shift. The branch-wise ratio $\mathrm{Re}(\omega)/[-\mathrm{Im}(\omega)] = 2Q$ shifts by -4.83%, -5.71%, and -7.31% for $n = 0, 1, 2$, respectively; the $n = 2$ value should be read with the overtone caveat already discussed. These ratio diagnostics and the fundamental $Q(a/M)$ curve are shown in Figure 5.

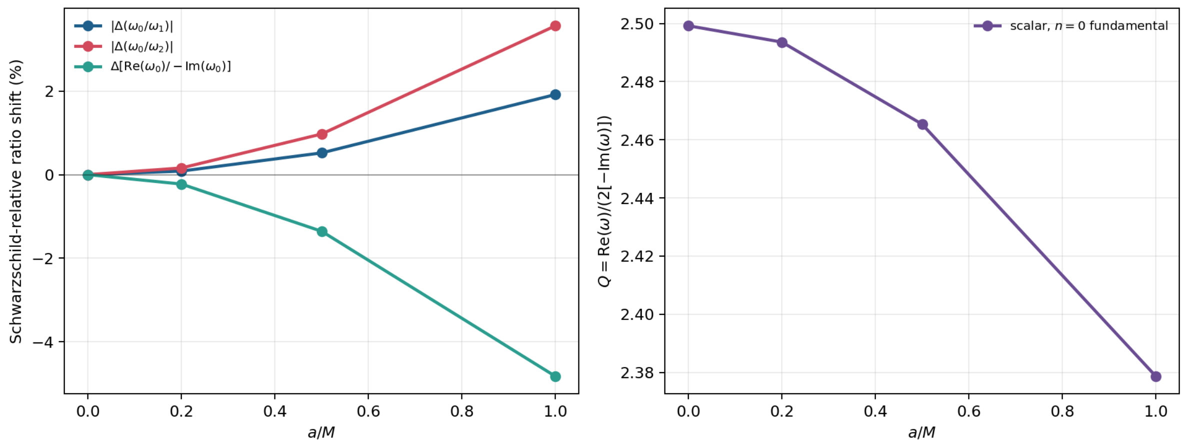

Figure 5 Scalar $\ell = 2$ dimensionless spectroscopy diagnostics. The left panel shows Schwarzschild-relative changes in ω_0/ω_1, ω_0/ω_2, and $2Q$. The right panel directly plots $Q = \mathrm{Re}\,\omega/[2(-\mathrm{Im}\,\omega)]$ for the robust fundamental branch. These quantities are catalogue diagnostics rather than an observational forecast.

This makes the scalar $\ell = 2$ fundamental branch the cleanest spectroscopic diagnostic in the present work. It is the most robust quadrupolar test-field branch in the present calculation, and its endpoint shift is large enough to be a meaningful catalogue-level effect. It is not the dominant gravitational-wave $\ell = m = 2$ ringdown mode. The ratio diagnostics strengthen the physics interpretation without turning it into an observational claim. The overtone shifts are also informative, especially for branch ratios and deformation sensitivity, but they should not be used as the primary claim until the $n = 2$ uncertainty and high-N branch tracking are tightened.

The trend should not be interpreted as a detection forecast. A uniform change in mass scale can mimic part of a frequency shift, so an observational spectroscopy analysis would need dimensionless ratios among modes, simultaneous use of damping information, and eventually a gauge-invariant gravitational perturbation treatment. The present result is therefore a catalogue diagnostic: across the sampled grid, KS deformation produces percent-level softening of the test-scalar QNM spectrum.

3.6 Comparison with existing KS literature

The present results should be read alongside, not as a replacement for, the existing KS QNM literature. Konoplya studied scalar, electromagnetic, and Dirac perturbations of the KS metric using WKB and time-domain methods, together with scattering and shadow diagnostics [16]. Bolokhov, Bronnikov, and Konoplya later emphasized that KS overtones can be considerably more sensitive to quantum deformation than the fundamental branch and used a Frobenius/continued-fraction treatment to study the overtone sector [17]. Those works establish that the KS deformation has observable spectral consequences in the model and that overtones require special care.

Direct row-by-row numerical comparison requires care because the tabulated conventions and parameter normalizations differ. In particular, much of the earlier KS QNM literature fixes the horizon radius, while the catalogue here is organized at fixed mass scale and reports a/M. For the lapse in Eq. (1), a horizon-normalized parameter $\beta = a/r_h$ corresponds to $\alpha = a/M = 2\beta/\sqrt{1-\beta^2}$, while $r_h\omega = \sqrt{4+\alpha^2}\,M\omega$. Thus the $a/M = 1$ endpoint in this paper corresponds to $a/r_h = 1/\sqrt{5}$, not to a horizon-normalized $a/r_h = 1$ table row.

A direct numerical overlap does exist for the scalar $\ell = 0, n = 0$ fundamental mode tabulated by Konoplya at fixed $r_h = 1$. This branch is outside the main $\ell = 2, 3, 4$ catalogue, so it is used only as a side diagnostic. After converting $\beta = a/r_h$ to $\alpha = a/M$, the same Chebyshev–Leaver workflow gives the horizon-scaled frequencies shown in Table 6. The agreement with the published time-domain values is at the 2.1×10^{-3}–2.6×10^{-3} relative level over the enabled rows; the comparison with the seventh-order WKB entries is weaker, roughly 9.4×10^{-3}–1.2×10^{-2}, as expected for low-ℓ WKB data. This check is useful because it shows that the present implementation reproduces the earlier fixed-horizon scalar fundamental branch within the accuracy of the published time-domain extraction, while the fixed-M catalogue answers a different normalization question.

$\beta = a/r_h$	$\alpha = a/M$	this work $r_h\omega$	Konoplya time-domain $r_h\omega$	$\Delta_{\rm rel}$
0.10	0.201008	0.221242 − 0.210676 i	0.221470 − 0.211440 i	2.603×10^{-3}
0.20	0.408248	0.222244 − 0.213397 i	0.222370 − 0.214200 i	2.631×10^{-3}
0.50	1.154701	0.229462 − 0.235795 i	0.230120 − 0.236200 i	2.343×10^{-3}

Table 6 Normalization-matched side comparison with the scalar $\ell = 0, n = 0$ fixed-horizon table of Konoplya [16]. The main catalogue in this paper starts at $\ell = 2$, so these rows are not used as headline results; they verify that the solver reproduces the previously tabulated scalar fundamental branch once the $r_h\omega$ convention is matched. The full six-row comparison is generated by `scripts/compare_konoplya2020_scalar_l0.py`.

The overtone comparison with Bolokhov–Bronnikov–Konoplya is necessarily more structural in the public manuscript record. Their overtone figures are for scalar $\ell = 0$ branches, including higher overtones $n = 5, 7, 9$, and the paper states that the precise numerical values plotted in those figures are available from the authors upon request [17]. The present catalogue instead reports $\ell = 2, 3, 4$ and stops at $n = 2$. The robust shared conclusion is therefore not a row-by-row frequency match, but the branch-status lesson: fundamentals are comparatively stable, while overtones are the deformation-sensitive sector and must be validated separately. Within the literature reviewed here, the distinctive contribution of the present work is the combination of fixed-mass KS catalogue generation, Leaver-anchored branch-status discipline, quality-factor shifts, and dimensionless spectroscopy diagnostics in a single spectral residual workflow.

3.7 Reported scalar rows

The scalar rows used for the headline numerical comparison are collected in Table 7. This table separates the baseline waveform diagnostics from the direct spectral values used for the paper's claims.

a/M	N	mode	status	$\mathrm{Re}(M\omega_{\mathrm{fit}})$	$\mathrm{Re}(M\omega_{\mathrm{pencil}})$	$\mathrm{Re}(M\omega_{\mathrm{spec}})$	$\mathrm{Im}(M\omega_{\mathrm{spec}})$
0.0	96	fundamental	high-N	0.48364248	0.48365353	0.48364387	-0.09675878
0.0	32	first overtone	cross-validated	–	–	0.46385058	-0.29560394
0.2	96	fundamental	high-N	0.48203239	0.48204305	0.48203369	-0.09665386
0.2	32	first overtone	cross-validated	–	–	0.46216559	-0.29531130
0.5	96	fundamental	high-N	0.47388656	0.47389459	0.47388693	-0.09610918
0.5	32	first overtone	cross-validated	–	–	0.45364275	-0.29379018
1.0	96	fundamental	high-N	0.44836165	0.44836213	0.44836241	-0.09424818
1.0	32	first overtone	cross-validated	–	–	0.42697149	-0.28857291

Table 7 Reported scalar comparison. Fundamental branches are reported at $N = 96$; first overtones are frozen at the Leaver-validated $N = 32$ reference grid. Displayed decimals are for branch identification and table readability; precision claims are set by convergence and Leaver validation, not by residual size alone. Tracked high-N overtone rows are retained as exploratory diagnostics rather than used as headline results.

3.8 Residual diagnostics

Table 8 reports scale-aware residual diagnostics associated with the same scalar rows. The polynomial backward error is

$$\eta_{\mathrm{back}}(\omega) = \frac{\sigma_{\min}[P_N(\omega)]}{\|A_0\|_2 + |\omega|\|A_1\|_2 + |\omega|^2\|A_2\|_2}. \tag{24}$$

a/M	N	mode	dim	$\sigma_{\min}(P_N)$	η_{back}
0.0	96	fundamental	96	1.036×10^{-19}	4.542×10^{-22}
0.0	32	first overtone	32	4.271×10^{-19}	1.713×10^{-20}
0.2	96	fundamental	96	8.244×10^{-21}	3.625×10^{-23}
0.2	32	first overtone	32	3.123×10^{-19}	1.256×10^{-20}
0.5	96	fundamental	96	5.098×10^{-20}	2.272×10^{-22}
0.5	32	first overtone	32	3.598×10^{-19}	1.468×10^{-20}
1.0	96	fundamental	96	4.017×10^{-20}	1.873×10^{-22}
1.0	32	first overtone	32	1.096×10^{-19}	4.696×10^{-21}

Table 8 Residual diagnostics for the reported spectral branches. Both quantities are evaluated by SVD on the raw, unscaled polynomial matrices; the row/column equilibration used to generate eigenvalue candidates is not used here. Backward error is scale aware but is still interpreted together with spectral-size and continued-fraction convergence.

Hermiticity and positive semidefiniteness of $R_N = P_N^\dagger P_N$ are algebraic properties, up to floating-point roundoff; they are not presented as empirical validation. The automated tests instead check their numerical implementation, Schwarzschild recovery, scaling invariance of the selected root, the revised axial potential, and the independent validation paths.

3.9 Local singular-value pseudospectral diagnostics of the scalar fundamental mode

The same residual operator gives a local finite-dimensional singular-value pseudospectral diagnostic around a validated branch. Following the singular-value viewpoint used in black-hole pseudospectrum studies [26, 27], define

$$\eta_N(\omega) = \frac{\sigma_{\min}(P_N(\omega))}{\sigma_{\max}(P_N(\omega))}. \tag{25}$$

This relative residual is used here only as a Chebyshev-matrix diagnostic for the scalar $\ell = 2, n = 0$ branch.

Figure 6 shows contours of $\log_{10} \eta_N$ around the residual-minimized scalar fundamental frequencies for $a/M = 0, 0.2, 0.5, 1.0$, using $N = 64$, an 81×81 grid, and a square half-width 0.025 in the complex $M\omega$ plane. The contour centers agree with the Leaver frequencies to at worst 6.816×10^{-12} in relative difference, so the plotted basins are attached to the validated scalar branch.

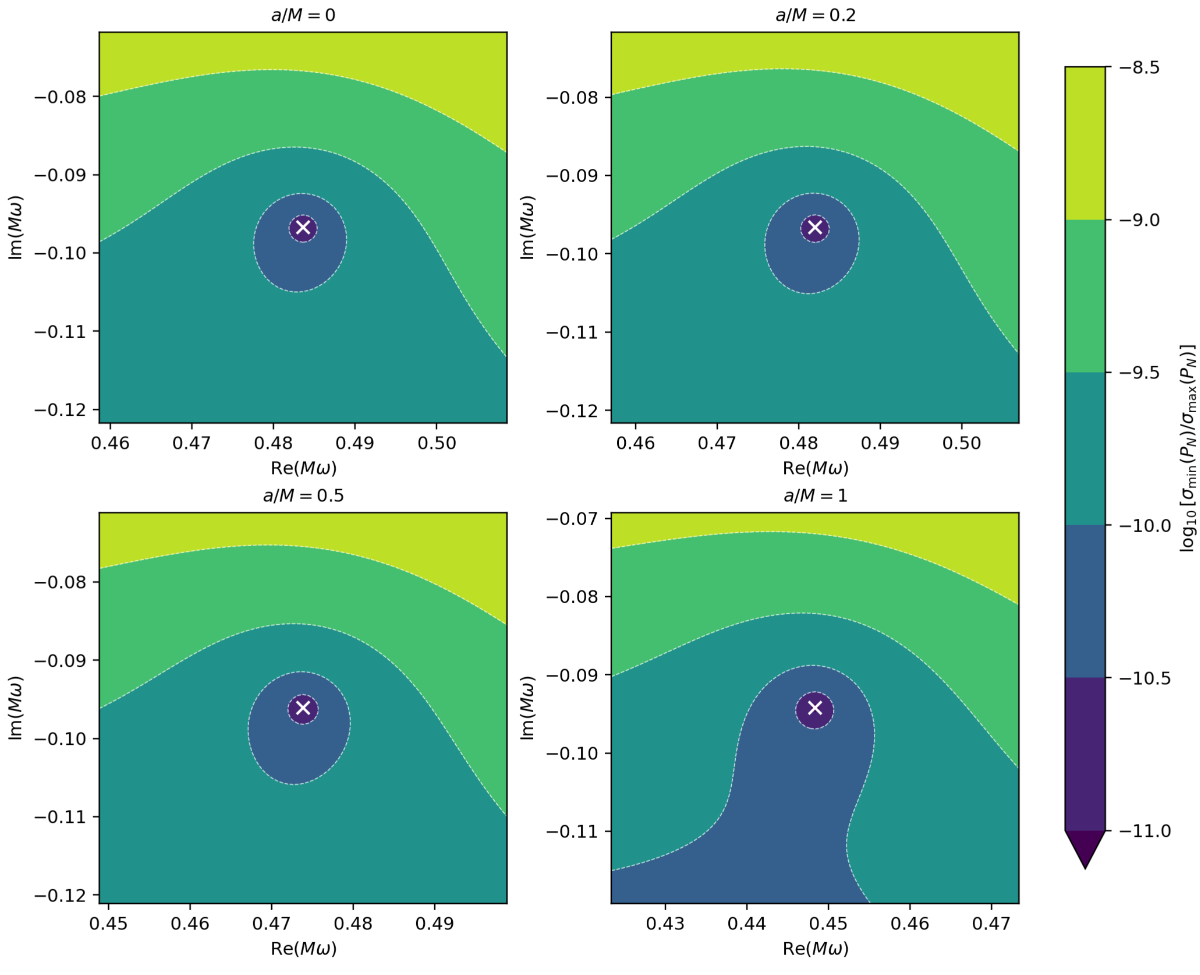

Figure 6 Finite-dimensional scalar $\ell = 2, n = 0$ singular-value pseudospectral contours for the relative residual $\eta_N = \sigma_{\min}(P_N)/\sigma_{\max}(P_N)$ at $N = 64$. White crosses mark the residual-minimized QNM locations. The panels show that the local small-η_N basin broadens as a/M increases.

Quantile and area diagnostics give the same trend. Let q_p denote the p-quantile of $\log_{10} \eta_N$ over the local window; lower-case q_p distinguishes this statistic from the QNM quality factor Q. At $N = 64$, $-q_{10}$ rises from 9.898 at $a/M = 0$ to 10.060 at $a/M = 1$, while $-q_{50}$ rises from 9.574 to 9.710. The area fraction satisfying $\log_{10} \eta_N \leq -10$ grows from 4.36% to 22.07%, a factor of 5.06. The upper-interval contour is window-limited, so the area is used only as a local diagnostic. These diagnostics are plotted in Figure 7.

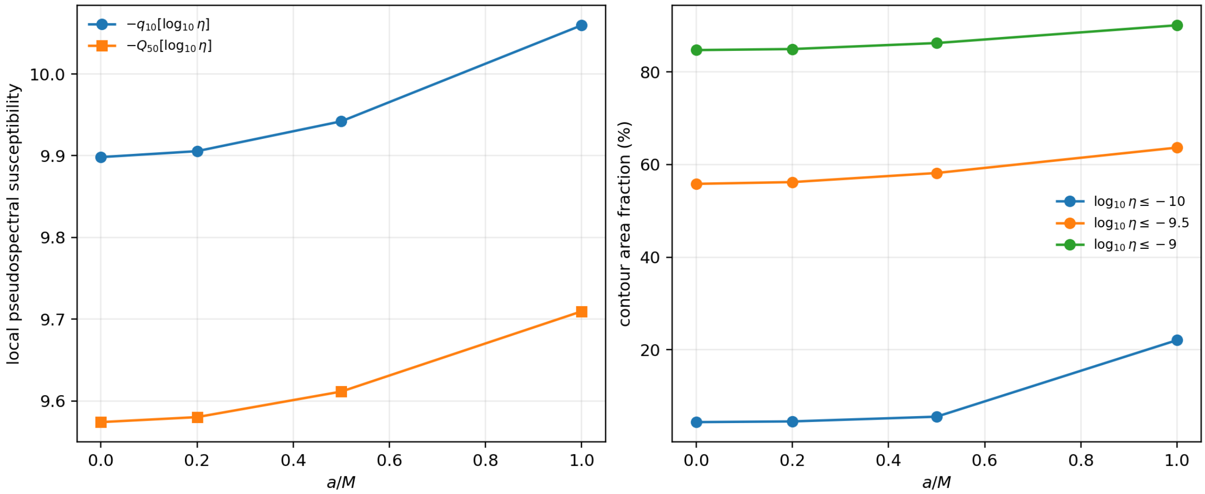

Figure 7 Scalar $\ell = 2, n = 0$ singular-value pseudospectral sensitivity diagnostics at $N = 64$. Left: quantile-based susceptibility measures increase with a/M. Right: fixed-threshold contour area fractions also grow; the deepest endpoint contour is window-limited and is used only as a local diagnostic.

The sign of the quantile trend is stable under the tested spectral sizes. As shown in Figure 8, the $a/M = 0$ to $a/M = 1$ gain in $-q_{10}$ is 0.097, 0.130, and 0.161 for $N = 32, 48, 64$, respectively. Thus, within the present residual normalization, scalar fundamental softening is associated with increasing local pseudospectral sensitivity.

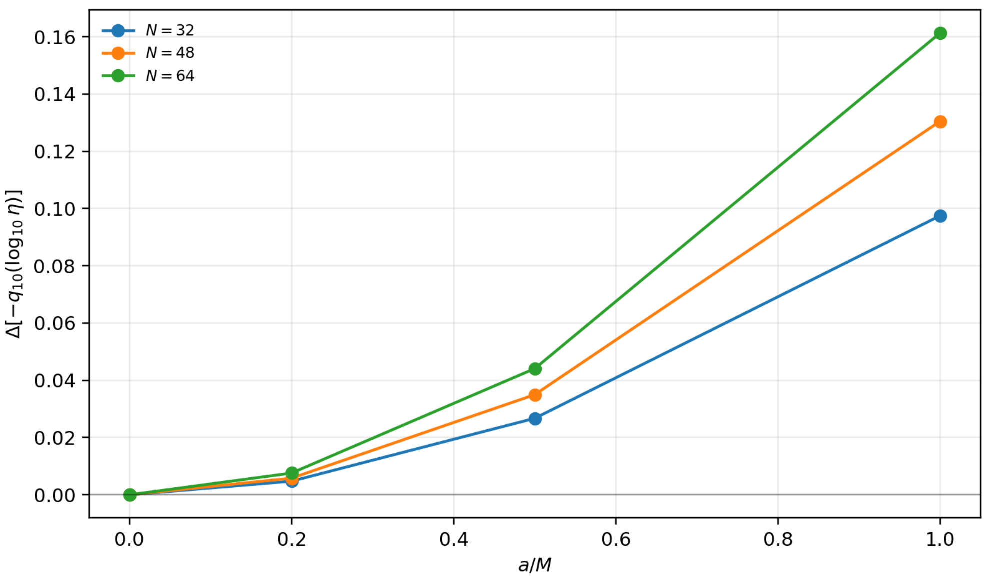

Figure 8 Finite-N check for the scalar $\ell = 2, n = 0$ singular-value pseudospectral trend. The plotted quantity is the increase in $-q_{10}(\log_{10}\eta_N)$ relative to the Schwarzschild value at the same N. The successive increase across the sampled a/M grid is stable for $N = 32, 48, 64$, although the absolute contour levels remain resolution dependent.

The grid, window, and spectral-size robustness audit is summarized in Table 9. It varies one numerical control at a time around the $N = 64$, 121^2-grid, half-width-0.025 reference. The endpoint change remains positive in all seven configurations and spans 0.1002–0.1647. The sign and order of magnitude are therefore robust, whereas the third decimal of a single-window result is not used as a physics claim.

test	N	grid	half-width	$\Delta[-q_{10}(\log_{10}\eta_N)]$
grid	64	81^2	0.025	0.1613
grid	64	121^2	0.025	0.1603
grid	64	161^2	0.025	0.1607
window	64	121^2	0.020	0.1396
window	64	121^2	0.030	0.1647
spectral size	32	121^2	0.025	0.1002
spectral size	48	121^2	0.025	0.1287

Table 9 Robustness of the scalar-fundamental endpoint susceptibility change between $a/M = 0$ and $a/M = 1$. Every configuration gives the same positive sign and comparable magnitude. Full-precision values are generated by `scripts/audit_pseudospectrum_robustness.py`.

4 Discussion

The central physical pattern is successive softening across the sampled KS grid. Increasing a/M across the four adopted values lowers both the oscillation frequency and the absolute damping rate for every sampled scalar and gauge-invariant inverse-Cowling axial branch in the catalogue. The scalar $\ell = 2$ fundamental mode is the least assumption-dependent interpretation of this pattern: at $a/M = 1$, its real frequency is lower by 7.29%, its damping magnitude is lower by 2.59%, and its quality factor is lower by 4.83%. The singular-value pseudospectral analysis adds a second numerical observation: in the same scalar branch, the local relative-singular-value basin broadens with KS deformation, consistent with greater local finite-N spectral sensitivity in the residual operator. This observation is intentionally local and exploratory.

The numerical framework is what makes that statement credible. The waveform-derived matrix-pencil path is retained only as a sanity check; the final operator is built directly from the perturbation equation. The Schwarzschild check shows that the direct spectral operator resolves the scalar fundamental mode accurately, and the Leaver comparison reproduces the targeted scalar spectral branches to better than 10^{-6} for the fundamental mode and first overtone at the reference grid. The low-lying catalogue extends this cross-check through $n = 2$; the largest observed spectral–Leaver difference is 1.202×10^{-5}, localized to a second-overtone axial branch. This confirms rather than erases the confidence hierarchy: scalar fundamentals are robust, scalar first overtones are cross-validated low-lying modes at $N = 32$, and second overtones are exploratory diagnostics; every axial tier remains conditional on the inverse-Cowling closure.

The workflow contribution is the KS-specific implementation, validation, and audit of a spectral residual workflow grounded in existing singular-value and pseudospectrum ideas. The generalized eigenproblem supplies candidates, but the accepted QNM claim is attached to a small value of $\sigma_{\min}(P_N(\omega))$, convergence in N, branch tracking, and Leaver comparison. This is a useful way to discuss the KS non-Hermitian resonance problem in Hermitian positive-semidefinite language, but it should not be read as a priority claim for residual singular-value theory.

4.1 Limitations and reproducibility

4.1.1 Interpretive boundaries

The scalar sector remains the cleanest physics sector because it follows directly from the test-field equation on the KS background. The axial metric variable is gauge invariant, and its equation follows from the general sourced odd-parity system only after an additional inverse-Cowling closure is imposed. Its interpretive boundary is therefore explicit: the invariant axial effective-source amplitudes are set to zero. Thus the axial entries are conditional odd-parity modes of this effective-source model, not a unique prediction of a still-unspecified quantum nonspherical completion.

The catalogue-level shifts are also not observational forecasts. The percent-level shifts indicate a physically meaningful deformation of the dimensionless spectrum, but an observational claim would require a waveform model, detector-noise weighting, parameter covariances, and a treatment of degeneracies with the remnant mass and spin. In particular, part of a frequency shift can be absorbed by a mass rescaling, and applying the axial result would additionally require accepting or replacing the stated effective-source closure.

The singular-value pseudospectral diagnostics are similarly bounded. They are computed from finite Chebyshev matrices $P_N(\omega)$, not from a rigorously controlled infinite-dimensional KS wave operator. The use of $\eta_N = \sigma_{\min}/\sigma_{\max}$ reduces simple matrix-scale effects but does not remove all discretization or norm choices. The fixed-threshold contour areas are therefore secondary diagnostics; the stronger statement is that the sampled quantile susceptibility increases successively for $N = 32, 48, 64$ on the scalar $\ell = 2, n = 0$ branch, and that its endpoint sign and order of magnitude survive the grid/window audit in Table 9. This should be read as an exploratory numerical observation attached to one validated scalar branch, not as a branch-independent statement.

Finally, the overtone hierarchy remains the numerically weakest part of the catalogue. Fundamental modes are the most stable, first overtones are reported quantitatively only on the Leaver-validated $N = 32$ grid, and tracked high-N first-overtone rows are retained as exploratory diagnostics. Second overtones are included because they are scientifically informative, but the largest spectral–Leaver difference occurs on an $n = 2$ branch. They should be interpreted as branch diagnostics until higher-precision continuation and additional validation are added; higher n is outside the scope. This claim hierarchy is summarized in Table 10.

Branch	Status	Use
Scalar $n = 0$	Robust fundamental modes	Main softening and spectroscopy claims.
Scalar $n = 1$	Cross-validated low-lying first overtones at $N = 32$	Secondary diagnostics; high-N tracking remains exploratory.
Overtones $n = 2$	Exploratory branch diagnostics	Avoid headline precision claims.
Axial rows	Conditional inverse-Cowling odd-parity modes	Gauge-invariant metric variable with frozen axial effective-source current.

Table 10 Claim hierarchy used in this manuscript. The table separates robust scalar fundamentals and cross-validated low-lying first overtones from exploratory second-overtone diagnostics and closure-dependent axial modes.

4.1.2 *Reproducibility*

The implementation is reproducible from the accompanying repository:

`https://github.com/adoolelomani2026/ks-qnm-spectral-residual-workflow`

The repository contains the Chebyshev spectral solver, the Leaver-style validation layer, catalogue-generation scripts, pseudospectrum diagnostics, generated tables, manuscript-supporting figures, and automated tests. The submission-facing commands are copied literally below.

```
pytest
python tests/test_qnm_algorithm.py --full
python scripts/run_hybrid_qnm_algorithm.py
python scripts/analyze_leaver_convergence.py
python scripts/analyze_solver_convergence.py
python scripts/analyze_scalar_potential_peaks.py
python scripts/audit_axial_model.py
python scripts/check_n128_spot.py
python scripts/analyze_pseudospectrum.py
python scripts/audit_pseudospectrum_robustness.py
python scripts/compare_konoplya2020_scalar_l0.py
```

The first two commands run the fast and complete scalar/axial validation suites. The remaining commands regenerate the main results, convergence audits, potential diagnostics, pseudospectrum data, and literature comparisons. These commands write the CSV tables, Markdown summaries, and figures under `outputs/results/` and `outputs/figures/`. The package versions used for the regenerated outputs are recorded in `outputs/results/run_metadata.json`.

5 Conclusion

The scalar sector supplies the physically controlled KS result. Its robust fundamental branches soften successively across the four sampled deformation values. For $\ell = 2$ at the upper sampled value $a/M = 1$, the oscillation frequency, damping magnitude, and quality factor shift by -7.29%, -2.59%, and -4.83%, respectively. Dimensionless scalar spectroscopy diagnostics shift by 1.92% in the cross-validated ω_0/ω_1 ratio; ratios involving $n = 2$ remain detailed exploratory diagnostics rather than headline results. The axial metric variable is gauge invariant and its potential differs explicitly from the earlier lapse-substitution ansatz. Its spectrum consists of conditional odd-parity modes within the inverse-Cowling closure, while a quantum treatment of independent nonspherical source excitations remains outside the spherical KS effective action and the present paper.

The numerical contribution is not a new high-precision eigensolver or a general solution of the high-overtone problem. It is the Chebyshev–Leaver audit framework behind the scalar statement: generalized eigenvalues are candidates, not conclusions, and are retained only after residual, spectral-size, continuation, backward-error, and collocation-independent continued-fraction checks. The continued-fraction truncation table makes those settings and their convergence explicit. The catalogue is deliberately low lying: $n = 0$ fundamentals are robust, scalar $n = 1$ modes are cross-validated low-lying first overtones at $N = 32$, $n = 2$ entries are exploratory branch diagnostics, and all axial modes are conditional inverse-Cowling odd-parity modes. Higher overtones are outside the scope.

The workflow contribution is the KS-specific implementation, validation, and audit of a spectral residual workflow grounded in existing singular-value and pseudospectrum ideas. The non-Hermitian KS resonance equation $P_N(\omega)\boldsymbol{u} = 0$ is tested through the Hermitian positive-semidefinite residual $R_N(\omega) = P_N(\omega)^\dagger P_N(\omega)$, and candidate modes are treated as minima of

$\sigma_{\min}(P_N(\omega))$ subject to convergence and continued-fraction validation. The singular-value pseudospectral diagnostic extends this workflow from locating frequencies to probing local finite-N spectral sensitivity: for the scalar $\ell = 2, n = 0$ branch, the sign and order of magnitude of the endpoint increase in the 10% quantile diagnostic survive changes in spectral size, grid density, and window width. The contribution is therefore not a claim of originality for residual singular-value theory or a continuum stability theorem. It is the combination of a fixed-mass KS Chebyshev residual implementation, continued-fraction validation, branch-status discipline, spectroscopy diagnostics, and a local finite-dimensional singular-value pseudospectral audit.

Acknowledgements and declarations

Funding. No funding was received to assist with the preparation of this manuscript.
Competing interests. The author declares no competing interests.
Data availability. The numerical data generated in this study, including catalogue rows, spectroscopy diagnostics, and scalar pseudospectrum grids, are available from the accompanying code repository and supplementary output files at `https://github.com/adoolelomani2026/ks-qnm-spectral-residual-workflow`.
Code availability. The code used to generate the spectral results, Leaver validation, catalogue tables, pseudospectrum diagnostics, and figures is available from the accompanying repository at `https://github.com/adoolelomani2026/ks-qnm-spectral-residual-workflow`.
Author contribution. Adel H. Al-Yoorby conceived the study, developed the numerical implementation, generated and analyzed the data, prepared the figures and tables, and wrote the manuscript.
Ethics approval. Not applicable.
Consent to participate. Not applicable.
Consent for publication. Not applicable.
Use of AI tools. OpenAI Codex with model identifier `gpt-5.6-sol`, as recorded in the relevant session metadata, was used during June–July 2026 for code development, numerical-analysis scripting, algebraic cross-checks, literature discovery, manuscript structure, language editing, and formatting. The author executed the calculations, checked cited claims against the primary sources, reviewed the derivations and numerical outputs, and takes responsibility for the final content.

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